Amlan Jyoti Sarmah

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Summary

Final-year Computer Science student at VIT Bhopal with hands-on experience developing full-stack web applications using React, Node.js, TypeScript, and Python. Skilled in REST API design, SQL and NoSQL databases, and machine learning model development using Scikit-learn. Also experienced in low-level C++ systems programming, having built a custom ML library from scratch. Self-driven and comfortable delivering end-to-end solutions independently.

Skills

- **Languages:** JavaScript, TypeScript, C++, Python, SQL
- **Frontend:** React.js, Next.js, Tailwind CSS, Socket.io, HTML, CSS
- **Backend:** Node.js, Express.js, Flask, REST API, Zod, WebSockets, Fast API
- **Databases:** PostgreSQL, MongoDB, ChromaDB
- **Machine Learning:** Scikit-learn, PyTorch, Pandas, NumPy, langchain
- **Tools:** Git, GitHub, Vercel, Linux

Education

- B.Tech, Computer Science and Engineering, VIT Bhopal University CGPA: 8.58 | Expected May 2027

Projects

Custom C++ Machine Learning Library ml-from-scratch-cpp.vercel.app | GitHub

- Implemented a lightweight, dependency-free machine learning library in C++ covering Linear Regression, Logistic Regression, K-Means Clustering, and Naive Bayes with a modular class-based architecture.
- Exposed model training and inference endpoints via a TypeScript API, interfacing directly with the custom C++ models.
- Authored a Python benchmark script to compare performance against Scikit-learn and visualized results on a React frontend deployed on Vercel.

Folio - Real-Time Collaborative Document Editor & AI Assistant github.com/AmlanJSarmah/Folio

- Architected a real-time collaborative markdown editor with React 19, TypeScript, and Socket.io, implementing dynamic room isolation and sub-100ms live event broadcasting.
- Engineered a distributed concurrency control mechanism using Redis atomic locks with heartbeat lease renewals and socket disconnect cleanup to prevent race conditions.
- Designed a read-through caching and write-buffering layer in Redis to minimize database load, persisting finalized documents to PostgreSQL via Prisma ORM upon lock release.
- Built a contextual Q&A chatbot microservice using a RAG pipeline with FastAPI, LangChain, and ChromaDB to perform semantic vector search over user notes.

Smart Tutor - Adaptive Learning System adaptive-learning-application.vercel.app | GitHub

- Developed an adaptive tutoring system that adjusts question difficulty in real time based on individual student performance.
- Trained a performance classifier using Scikit-learn and deployed it via Flask as a REST API for real-time difficulty predictions.
- Engineered a separate backend service with TypeScript, Express, and Zod to handle authentication, question recommendation, and request validation.
- Modelled student profiles, performance records, and question banks in MongoDB to power the recommendation engine.

Work Experience

Technical Team Lead, North East Club, VIT Bhopal *August 2024 to August 2025*

- Led a team of 11 developers, driving all technical decisions for a club of 100 plus members.
- Designed and deployed the club's official responsive website using React.js and Tailwind CSS, improving information accessibility for 100 plus members.
- Coordinated technical logistics and on-site support for 3 club events, each attended by 150 to 200 participants.